

DUSTBADGE · CALCULATION

DustBadge sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	First issue for TRL 3 (power and run time, charging, badge temperature, PM4 against the respirable convention, range, error budget, alert logic, log, breathing zone, mass, drop, cost)

DustBadge sizing calculations

On paper, DustBadge meets nine of its fifteen requirements (five by calculation, four by design), has one at risk, cannot verify one at TRL 3 and misses four. The misses are the working range (R2: the sensor stops at 1 mg/m³ against a 5 mg/m³ target), accuracy after site calibration (R3: about ± 34 % for quartz-rich stone at the action level, because an 8 h filter sample collects only 0.037 mg), operation in full sun at 45 °C (R11: the shell reaches about 63 °C, above the sensor's 60 °C limit) and intrinsic safety (R14, out of scope). Run time (R7) is at risk: 13.3 h at the sensor's typical current, but 11.3 h at its maximum current or at 0 °C. The calculations changed two parts of the TRL 2 concept: a humidity and temperature sensor is added to the carrier board so that R11's flag can work at all, and the charger takes a cell thermistor so that the 0 to 45 °C charging rule is enforced. Both add \$3, and the parts now cost \$88 against the \$90 budget. Every number in this note is printed by `docs/04-calcs/sizing.py`; the tag in brackets, for example [A3], is the line of that script's output that carries it.

Safety: These are first-principles estimates for a paper proof of concept. They do not show that a worker's air is safe, and they are not a substitute for co-located filter sampling, chamber checks or electrical safety checks on the lithium cell. See DBG-PRC-001, Safety.

■ Scope and method

The note checks every requirement in DBG-REQ-001 v0.3 against the design in DBG-PRC-001 v0.3 and the parametric model `cad/src/model.py`. The script imports the model's `PARAMS`, derived dimensions and mass breakdown, so the envelope, shell volumes, inlet position and cell space used here are the ones in the STEP files and in drawing DBG-DWG-001. It also reads `bom/bom.csv` and `budget_usd` in `project.yaml`. Run it from the repo root with `python docs/04-calcs/sizing.py`.

The design case is a worker in a surface quarry or a stone fabrication shop on a 12 h shift, badge worn at the collar or on an upper harness strap, ambient 0 to 45 °C, 10 to 90 % RH, US OSHA and MSHA limits by default (decision D3 in DBG-DDR-001).

■ Assumptions

Table 1. Main assumptions.

Area	Assumption	Basis
Particle sensor	45 to 55 mA typical and 65 mA maximum at 4.5 to 5.5 V; 0 to 1,000 $\mu\text{g}/\text{m}^3$; PM4 precision $\pm 25 \mu\text{g}/\text{m}^3$ below 100 $\mu\text{g}/\text{m}^3$ and $\pm 25 \%$ from 100 to 1,000 $\mu\text{g}/\text{m}^3$; drift 1.25 $\mu\text{g}/\text{m}^3$ or 1.25 % per year; best performance at 10 to 40 °C and 20 to 80 % RH; operating -10 to 60 °C and 0 to 95 % RH; new readings every second; 26.3 g	Sensirion SPS30 datasheet, v2.0, June 2023
Electronics	5 V boost 85 % efficient; controller, BLE and humidity sensor 3 mA from the cell; vibration motor 80 mA at 3 V, 20 alerts of 2 s per shift	Typical module figures; to confirm by measurement at a later TRL
Cell	1,500 mAh, 3.7 V; 80 % usable; 85 % of capacity at 0 °C; charged at 500 mA	Typical LiPo data
Badge temperature	Combined surface coefficient 10 W/(m ² ·K); solar absorptance of yellow PETG 0.45; 1,000 W/m ² on the front face; rear face against the wearer treated as adiabatic	Screening values (assumed)
Reference sample	Cyclone at 2.5 L/min for 8 h; weighing and blank uncertainty 0.010 mg per filter; gravimetric method $\pm 10 \%$	Assumed typical values
Error terms	Site factor change between days and tasks $\pm 10 \%$ (factor per task) or $\pm 20 \%$ (per site); position and screen $\pm 10 \%$; humidity below 80 % RH $\pm 5 \%$; non-linearity after the site factor $\pm 10 \%$; badge-to-badge after a CalRig check $\pm 10 \%$	Assumed; each to be measured
Silica fractions	10 % for a typical quarry, 80 % for quartz-rich engineered stone	Assumed for illustration; a site filter sample sets the real value

■ A. Power and run time (R7)

The cell holds 5.55 Wh, of which 4.44 Wh is usable, and 3.77 Wh at 0 °C [A1]. The motor adds only 2.7 mWh per shift [A2]. The sensor's fan and laser dominate.

Table 2. Run time by sensor current [A3].

Sensor current	Power from the cell	Run time at 25 °C	Run time at 0 °C
45 mA (typical, low)	276 mW	16.1 h	13.7 h
55 mA (typical, high)	335 mW	13.3 h	11.3 h
65 mA (maximum)	393 mW	11.3 h	9.6 h

The 12 h target is met at typical current and room temperature and missed at the maximum current or at 0 °C with the upper typical current. Covering the worst case needs about 1,880 mAh [A4], or about 1,600 mAh at 25 °C [A5]. The model leaves 1.5 mm behind the cell, so a cell up to 11.5 mm thick fits without changing the shells [A6]. A larger cell is proposed in `docs/REVIEW.md`, awaiting Amish, because it adds about 8 g (R9) and about \$3 (R15).

■ B. Charging

At 500 mA (0.33 C) the cell charges in about 3.4 h, 2.4 h at constant current plus about 1 h of taper [B1], so a badge charged after a shift is ready for the next. The charger module now takes the cell's thermistor so that charging stops outside 0 to 45 °C; at TRL 2 this rule was written but nothing enforced it.

■ C. Badge temperature (R11)

In shade the electronics warm the badge by 3.3 K. In full sun on the front face the rise is 17.8 K: the shell reaches about 48 °C at 30 °C ambient and about 63 °C at 45 °C ambient [C1]. The sensor's 60 °C operating limit is therefore passed in full sun above about 42 °C ambient, and its 40 °C best-performance limit above about 22 °C ambient [C2]. The sensor draws outside air through its own path, so the air it measures stays near ambient, but its electronics and the cell sit at shell temperature. R11 is not met in full sun at 45 °C. A lighter shell color or a vented sun hood would help; neither is adopted here.

■ D. PM4 against the respirable convention (R1)

The ISO 7708 respirable convention passes half of particles at 4.0 µm aerodynamic diameter [D1]. For quartz (2.65 g/cm³), geometric and aerodynamic diameters differ by a factor of about 1.6: a 2.5 µm grain is 4.1 µm aerodynamic and about half respirable, and a 4 µm grain is 6.5 µm aerodynamic and only 12 % respirable [D2]. An optical PM4 channel would therefore over-include coarse quartz if it sized it correctly. In practice the SPS30 calculates PM4 from the distribution of the particles it detects best, which are fine, so it tends to under-see coarse grains. The two errors act in opposite directions and neither is known for site dust [D3]. PM4 remains the closest available output, so R1 is met by design as a proxy, and the site factor must absorb the difference.

■ E. Working range (R2)

The sensor covers 0 to 1,000 µg/m³, 20 % of the 5 mg/m³ target span [E1]. What matters for the alert is where the silica limits fall in dust terms.

Table 3. Respirable dust at the RCS action level and limit, by silica fraction [E2].

Silica fraction	Dust at the 25 µg/m ³ action level	Dust at the 50 µg/m ³ limit	Limit in range
2.5 %	1,000 µg/m ³	2,000 µg/m ³	No
5 %	500 µg/m ³	1,000 µg/m ³	Yes, at the edge
10 %	250 µg/m ³	500 µg/m ³	Yes
30 %	83 µg/m ³	167 µg/m ³	Yes
80 %	31 µg/m ³	62 µg/m ³	Yes

For dust with 5 % silica or more the limit lies inside the range, and for 2.5 % or more the action level does [E3]. Above the range, readings are not trustworthy, so the firmware logs an over-range minute at the range limit, flags it and counts it toward the projection at no less than that limit, which makes the badge alert rather than go quiet [E4]. R2 is not met: peaks in quarries and artisanal mines can be far above 1 mg/m³, and the badge will report only that they were over range.

■ F. Error budget for the shift average (R3)

A site factor from a co-located filter sample is what turns the optical reading into a respirable dust value. The filter sample itself limits the result at low concentrations: 2.5 L/min for 8 h draws 1.20 m³ of air [F1], so at the action level for quartz-rich stone (31 µg/m³ of dust) the filter collects only 0.037 mg, and a 0.010 mg weighing uncertainty is already 27 %.

Table 4. Combined uncertainty of the shift average at the action level (root sum of squares) [F2, F3].

Case	Dust	Filter mass	Same badge, factor per task	Factor shared across badges, per site
Quartz-rich stone, 80 % silica	31 $\mu\text{g}/\text{m}^3$	0.037 mg	$\pm 34 \%$	$\pm 39 \%$
Quarry, 10 % silica	250 $\mu\text{g}/\text{m}^3$	0.300 mg	$\pm 21 \%$	$\pm 29 \%$
Quartz-rich stone, 4.2 L/min cyclone over 2 shifts	31 $\mu\text{g}/\text{m}^3$	0.126 mg	$\pm 22 \%$	

The $\pm 25 \%$ target is met only for the quarry case with the reference taken on the same badge for each task, and for quartz-rich stone only with a larger reference sample [F3]. Without any site factor the PM4 precision alone is $\pm 25 \mu\text{g}/\text{m}^3$, 80 % of the dust level at the action level for quartz-rich stone [F4]. R3 is not met on paper for the general case. Every term except the sensor's own figures is assumed and must be measured.

■ G. Silica estimate and alert logic (R4, R5, R6)

The sensor gives new readings every second; the firmware logs 1 min means, multiplies calibrated PM4 by the site silica fraction to give the RCS estimate, and projects the 8 h TWA as the dose so far plus the last 60 min mean rate for the rest of the shift. The projection starts once 60 min of data exist; before that the badge alerts only on the dose so far and on peaks. A brief peak alert fires when a 1 min mean exceeds 250 $\mu\text{g}/\text{m}^3$ RCS, ten times the action level [G2]. Five scenarios on an 8 h shift test the logic [G1].

Table 5. Alert behavior in five scenarios (RCS, $\mu\text{g}/\text{m}^3$) [G1].

Scenario	8 h TWA	Action-level alert	Limit alert	Peak alerts	Correct?
S1 steady 30 all shift	30.0	Minute 60	None	0	Yes
S2 background 10, 2 min dry cut at 1,000 each hour	43.0	Minute 60	None	8	Yes
S3 background 15, one 1 min truck pass at 110	15.2	None	None	0	Yes, no nuisance alert
S4 background 5	5.0	None	None	0	Yes
S5 background 10, one 2 min dry cut at 1,000	14.1	Minute 60	None	1	Early warning; the shift ends below the action level

The logic warns before the limit in every case where the shift ends above the action level, and it ignores a passing truck. S5 shows the cost of projecting: a single heavy cut in the first hour projects over the action level although the shift ends at 14 $\mu\text{g}/\text{m}^3$. That is acceptable for a warning, since the same cut repeated each hour gives S2, but workers should be told what the alert means. R4, R5 and R6 are met by design; R6 is also met on paper for these scenarios.

■ H. Shift log (R12)

At 16 bytes per minute a 12 h shift takes 11.5 kB, and 30 shifts take 346 kB, 16 % of the module's 2 MB flash, so the badge holds about 182 shifts [H1]. At a conservative 2 kB/s a shift transfers to a phone in about 6 s [H2]. R12 is met on paper.

■ I. Breathing zone (R8)

With the badge worn 210 mm below and 70 mm to the side of the nose and mouth, the inlet is 242 mm from them [I1]. The badge center may sit up to 269 mm below the nose and mouth before the 300 mm limit is passed [I2], so a collar or upper-strap position meets R8 and a low shirt pocket may not. R8 is met on paper for the stated mount.

■ J. Size and mass (R9)

The printed shells weigh 16.5 g and 15.1 g from the model volumes; the sensor (26.3 g) and the cell (30 g) are more than half the badge [J1]. The total is 111.8 g, 8.2 g under the 120 g target, and the envelope is 64 x 52 x 33 mm including the clip against 75 x 55 x 35 mm [J2]. A cell 8 g heavier would leave 0.2 g of margin [J3]. R9 is met on paper.

■ K. Drop and ingress (R10)

A 1.5 m drop reaches 5.42 m/s with 1.65 J. If the shell corner crushes 1 to 2 mm, the mean deceleration is 750 to 1,500 g, so the sensor needs 194 to 387 N of retention and the cell 221 to 441 N [K1]. The model holds the sensor between the front shell floor and the carrier board, which rests on three screw bosses; whether PETG bosses and a foam-backed cell survive these loads, and whether the gasketed joint reaches IP54 for the electronics, cannot be shown without a test. R10 is not verifiable at TRL 3.

■ L. Cost (R15)

The BOM has 12 lines totaling \$88.00 against `budget_usd` of \$90, a \$2.00 margin; the particle sensor is 55 % of the cost [L1]. R15 is met on paper, with little room for the larger cell in section A.

■ M. Results against every requirement

Table 6. Requirement status from this note.

ID	Requirement	Value	Target	Status
R2	Cover the working range	0 to 1,000 µg/m ³ ; limit in range for silica fractions of 5 % or more; over-range flagged and counted	0 to 5 mg/m ³	Not met
R3	Accuracy after site calibration	±34 % (quartz-rich stone), ±21 to ±29 % (quarry); ±22 % with a larger reference sample	±25 % of a gravimetric shift average	Not met on paper (assumed terms)
R11	Work in site conditions	Shell about 63 °C in full sun at 45 °C; sensor best performance only to 40 °C and 80 % RH; humidity flag now possible	0 to 45 °C, 10 to 90 % RH, readings flagged	Not met in full sun above 42 °C
R14	Safe in hazardous atmospheres	Not intrinsically safe	Certified intrinsically safe	Not met , out of scope
R7	Last a full shift	13.3 h typical at 25 °C; 11.3 h at maximum current or at 0 °C	12 h continuous	At risk
R10	Survive the site	750 to 1,500 g on a 1.5 m drop; IP54 by gasket, untested	IP54 electronics; 1.5 m drop	Not verifiable at TRL 3
R6	Warn before a limit is reached	Five scenarios behave as intended; one early warning (S5)	Projected TWA alerts at 25 and 50 µg/m ³	Met on paper
R8	Wearable in the breathing zone	Inlet 242 mm from nose and mouth	300 mm or less	Met on paper
R9	Light and small	111.8 g; 64 x 52 x 33 mm	120 g; 75 x 55 x 35 mm	Met on paper
R12	Keep a shift log	11.5 kB per shift; 182 shifts; 6 s transfer	30 shifts; BLE export	Met on paper
R15	Low cost and buildable	\$88.00; perfboard carrier	\$90; no custom PCB	Met on paper
R1	Measure the respirable fraction	PM4 as proxy; size cut offset in both directions (section D)	PM4 as ISO 7708 proxy	Met by design
R4	Estimate silica exposure	Calibrated PM4 x site fraction, labeled	As stated	Met by design
R5	Update often enough	1 s readings, 1 min log, TWA every minute	As stated	Met by design
R13	Protect the worker's data	Badge and phone only; sharing by the worker	As stated	Met by design

Counts: 4 not met, 1 at risk, 1 not verifiable at TRL 3, 5 met on paper, 4 met by design.

■ Checks against the TRL 2 figures

Table 7. TRL 2 claims checked.

TRL 2 claim (DBG-PRC-001 v0.2)	This note	Action
About 335 mW from the cell	335 mW at 55 mA; 393 mW at 65 mA	Stands
About 13 h typical, about 11 h worst case	13.3 h and 11.3 h; also 11.3 h at 0 °C	Precis updated with the cold case
Charge about 3.5 h	3.4 h	Stands
Sensor precision $\pm 5 \mu\text{g}/\text{m}^3$ plus 5 % at the action level for quartz-rich stone	That is the PM2.5 figure; PM4 is $\pm 25 \mu\text{g}/\text{m}^3$ below 100 $\mu\text{g}/\text{m}^3$	Precis corrected
Log about 12 kB per shift, 30 shifts about 350 kB	11.5 kB and 346 kB	Stands
About 220 mm from badge to nose	Inlet 242 mm from the nose and mouth midpoint	Precis and requirements updated
About 110 g	111.8 g	Precis updated
64 x 52 x 30 mm	64 x 52 x 30 mm shells, 33 mm with the clip	Precis updated
About \$85	\$88.00 with the humidity sensor and thermistor charger	Precis, BOM notes and README updated
Badge temperature not estimated	About 63 °C shell in full sun at 45 °C	R11 status changed to not met